GEK-73809A

DRIVE SYSTEM INSTRUCTIONS

DC REGULATOR CONTROLS

For

SCR EXCITATION SYSTEMS

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

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These instructions are to be used as a guide for installing, operating and maintaining the options on the SCR excitation system supplied on MS5001, MS6001 and MS7001 gas turbine driven generator.

Only personnel that are adequately trained and are thoroughly familiar with the equipment and these instructions should install, operate and maintain this equipment.

These instructions are to be used in conjunction with the excitation system instructions and the installation instructions.

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* *
* CAUTION *
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THERE IS A POSSIBILITY OF CONTROL MALFUNCTION IF A RADIO TRANSMITTER IS KEYED IN THE VICINITY OF PROCESS CONTROL OR POWER CONVERSION EQUIPMENT. THE INTERFERENCE MAY BE ESPECIALLY NOTICEABLE WITHIN FOUR FEET OF EQUIPMENT WITH PANEL DOORS OPEN.

It is recommended that walkie talkie type radios not be used in close proximity of operating electronic equipment.

Keeping panel doors closed and having proper frame to ground grid system connections will minimize the problem but may not eliminate it.

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* *
* WARNING *
* *
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THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

Throughout this instruction book there are two conventions used for highlighting names. Abbreviations, signal names, names of adjustable devices and test points are capitalized as in the following examples: PT, VREF, 2P (TEST), 6TP. However, some device names also capitalized, such as 59EX, 1T, 1VR, etc. The names of modules, submodules and printed wiring boards (cards) are underlined when they occur within the body of the text. Examples of this type of highlight are as follows: Signal Conditioner module, Auto Regulator PWB, AC/DC PWB.

RECEIVING AND STORAGE

As soon as the equipment is received, it should be examined for any damage that might have been sustained in transit and all shipping material removed. If injury or rough handling is evident, a damage claim shall be filed immediately with the carrier and the nearest General Electric sales office should be notified promptly.

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If the equipment is to be stored the space heaters must be energized to prevent moisture condensation.

ABBREVIATIONS

The following abbreviations are used in this instruction book.

AC, ac Alternating Current

AFFL Amps, Field, Full Load

AFNL Amps, Field, No Load

ARCC Active and Reactive Current Compensation

CT Current Transformer

D/A Digital to Analog

DC, dc Direct Current

GAC Generator Auxiliary Compartment

GEC Generator Excitation Compartment

I Current

IC Integrated Circuit

LDC Line Drop Compensator

LED Light Emitting Diode

LSB Least Significant Bit

MAX Maximum

MIN Minimum

MW Megawatts

PCB Printed Circuit Board (same as PWB)

PB Pushbutton

PF, pf Power Factor

PPT Power Potential Transformer

PS Power Supply

PSS Power System Stabilizer

PT Potential Transformer

PU Per Unit

PWB Printed Wiring Board (same as PCB)

RC Resistor Capacitor

RCC Reactive Current Compensation

SCR Silicon Controlled Rectifier

SVA Static Voltage Adjuster

TP, Test Point

TVM Transfer Voltmeter

URAL Under-excited Reactive Ampere Limit

V Volts

VARS Volt Amperes, Reactive

VFFL Volts, Field, Full Load

VFNL Volts, Field, No Load

V/Hz Volts/Hertz

INTRODUCTION

The DC Regulator Controls described in this instruction book are provided as an option on static excitation systems for gas turbine systems. The DC Regulator Controls provide manual operator control as well as automatic control of the DC Regulator on-line setpoint.

The regulator circuitry of the static excitation system consists of two basic regulators: an AC Regulator and a DC Regulator. The AC Regulator is designed to regulate generator terminal voltage. The DC Regulator is designed to regulate generator field volts. The DC Regulator is part of the inner loop regulator, and when the DC Regulator mode is selected, the DC setpoint controls the field voltage. When the AC Regulator mode is selected, the AC Regulator controls the setpoint of the inner loop. When a higher or lower generator terminal voltage is required, the AC Regulator automatically makes adjustment to the inner loop, which will vary the field voltage, which in turn will vary the terminal voltage. The DC Regulator, when not provided with the DC Regulator control option, has two fixed DC setpoints: the off-line setpoint and the on-line setpoint. These setpoints are determined by the settings of 90R5 and 90R6 potentiometers respectively. Under normal operating conditions, these potentiometers are fixed and are not accessible for adjustment. The off-line DC Regulator set-point provides a no-load field voltage rating for the system on start-up and shut-down. The on-line DC set point, determined by 90R6, is normally set at 75% of full load field voltage rating. This set-point is usually a safe value of excitation for most system operating conditions. In other words, it is high enough so that a loss of synchronism does not occur if the DC Regulator were to gain control.

In the rare occasion of a failure of the AC Regulator circuits, an automatic transfer from AC Regulator to DC Regulator occurs. Since the DC Regulator setpoint is fixed, the operator has no method to control the generator voltage. The DC Regulator control option provides the operator with the means to control the terminal voltage manually while on line. In addition, the DC Regulator is provided with the Automatic Tracking feature, which causes the DC Regulator setpoint to follow or track the AC Regulator output while in the AC Regulator mode. This feature allows a transfer to be made from AC to DC Regulators without a change in excitation. To give the operator a visual means to make a transfer from DC to AC, a Transfer Voltmeter is provided which displays the difference between the AC and DC Regulator outputs. In addition, a light is provided which informs the operator which control mode controls the generator voltage.

DESCRIPTION

The DC Regulator Control consists of several printed wiring boards (PWB) as shown in Figure 1. Included in the option are the DC SVA Control PWB and the DC SVA Counter PWB and the related remote Raise and Lower controls. Also included is the Transfer and Tracking PWB to provide the automatic tracking function. The Volts to Milliamp Converter is required to drive the Transfer Voltmeter at the Generator Control Panel.

An indication of which regulator mode the system is operating in is provided for the operator. This is provided through a relay driver and relay, 83RM, and lights to indicate control by the AC or DC Regulators. The Exciter Cubicle Alarm PWB is used to drive the 83RM relay.

PRINCIPLES OF OPERATION

The function of the DC Regulator Control is to provide an on-line DC Regulator set point that will always be equal to the AC Regulator setpoint when operating on-line in AC Regulator mode and to be able to adjust the DC setpoint manually when operating on-line in the DC Regulator mode.The purpose of this is to provide a smooth response in excitation when a switch from AC Regulator to DC Regulator occurs. Any difference in the setpoints of the regulators may cause an abrupt change in the generator field voltage which may induce a change in generator terminal volts and VARS.

The DC Regulator Control option is centered around the DC SVA. This is the circuitry that accepts the Raise and Lower input signals from the DC Regulator set point adjuster and outputs the DC Regulator setpoint to the inner loop. This setpoint is adjustable from remote locations as well as from a local control. The DC SVA consists of two PWBs, the SVA Control and the SVA Counter PWBs. The SVA Control PWB accepts the various controlling inputs, and by proper logic, commands the SVA Counter PWB. This PWB is an up-down digital counter with a digital-to-analog (D/A) converter connected to its output. The output of the D/A converter is used as the reference for the DC regulator.

The SVA Control PWB accepts the following input signals for control of the raise and lower signals (See Figure 2):

1. Operator Control (OP). Contact for raise and lower are input by the operator. If both are closed simultaneously, the raise will be selected. These control the analog output of the SVA Counter (if the automatic operation is not selected). For the standard GT application of the DC SVA, 90MR and 90ML are the operator raise and lower relays.

2. Automatic Operation (AUTO). This is a contact which, when closed, blocks the operator inputs from effecting the raise or lower control. This input is used for auto tracking.

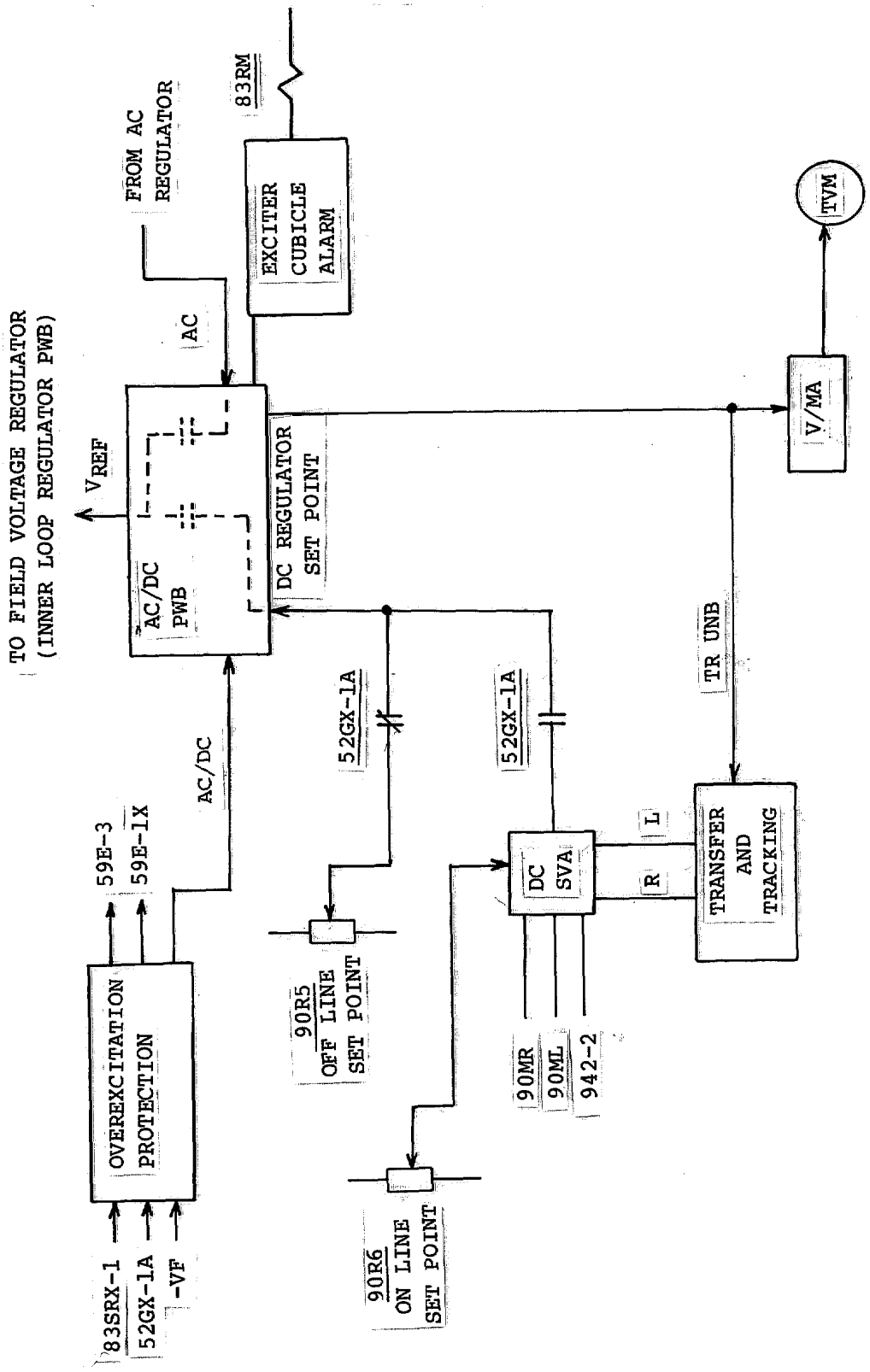


Figure 1
DC Regulator Adjust

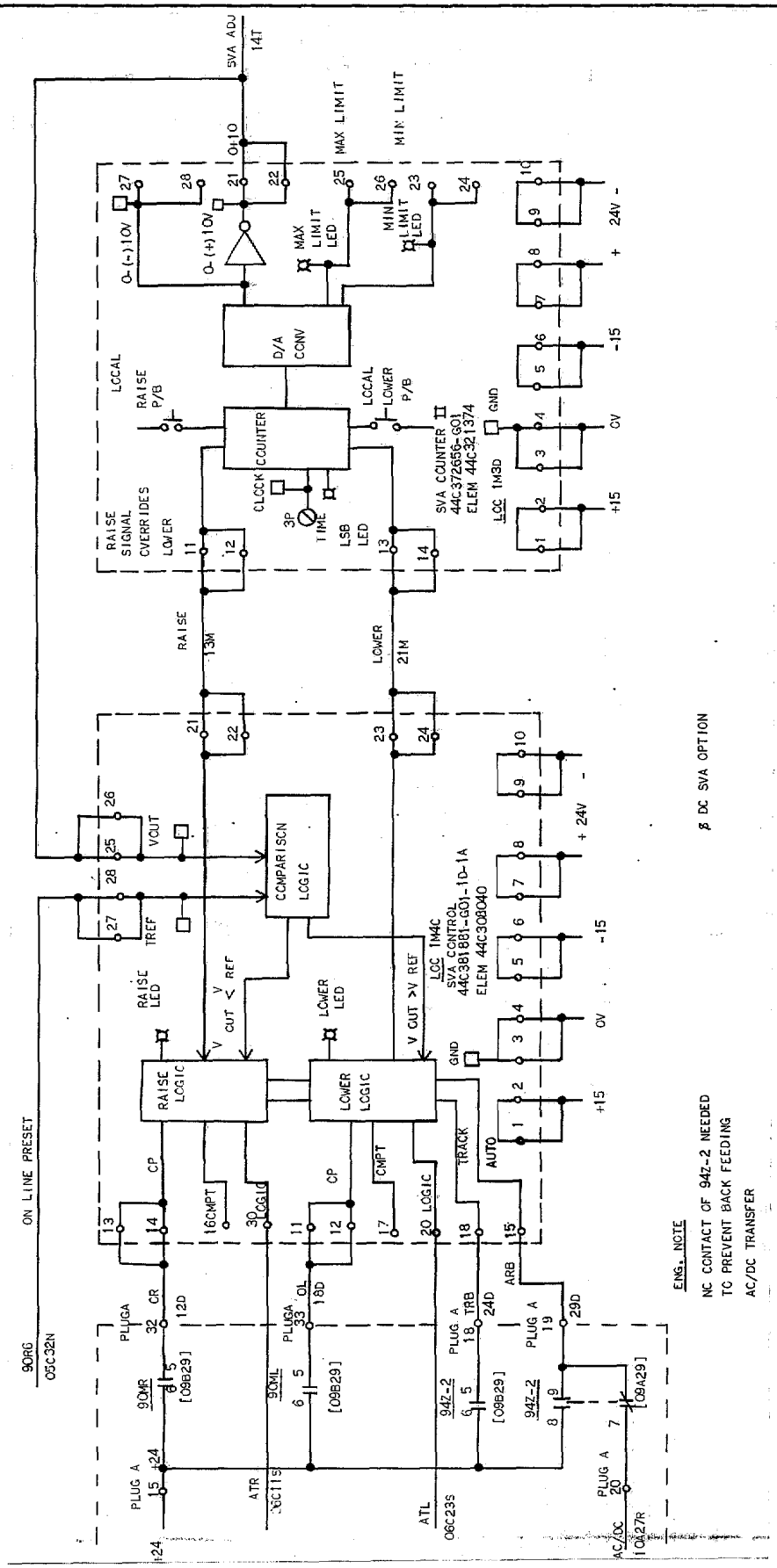


Figure 2
DC Static Voltage Adjustment Circuitry

3. Computer Control (CMPT). A computer or some other automatic device may cause contacts to raise or lower the SVA output. They will have control only when the automatic operation contact is closed. They are not used in this application.

4. Logic Control (Logic). Logic signal inputs may also raise or lower the SVA output voltage. They are enabled by the closure of the automatic operation contact. They are used by the auto tracking function.

5. Tracking (Track). When the SVA output signal is to follow an analog input reference, an input contact closure is required. When this contact opens, tracking stops immediately.

6. Analog Inputs. Two inputs are available for analog signals. They are compared to determine whether to raise or lower the SVA output voltage. Both the tracking and the automatic operation contacts must be closed for these inputs to control the SVA output voltage.

The SVA Counter PWB will count down if there is a logic 1 on the lower input (pins 11,12) and will count up if there is a logic 1 on the raise input (pins 13,14). In addition, there is a raise and lower pushbutton on the faceplate of the PWB that will also control the counter. For any inputs, the raise input will always override and cause the counter to count up. The rate at which the counter counts up or down is determined by an oscillator, whose frequency is adjustable by potentiometer 3P. The range of oscillator frequencies (Clock TP) is from 33 to 1250 Hz (.8 millisecond to 30 milliseconds). This results in a time from minimum to maximum limit of 5 to 125 seconds (4096 clock cycles). The nominal factory set time is 120 seconds.

Connected to the ten most significant bits of the counter is a D/A converter. The analog output is from 0 to +10 VDC.

In addition to the analog output, there is a local indicator (LEDs) and logic output for minimum and maximum limits.

During a start-up of a gas turbine generator, the inner loop regulator requires a reference to set the field voltage level to VFNL. This reference is provided by the DC off-line setpoint, which is the fixed voltage setting provided by potentiometer 90R5 shown in Figure 3. The DC off-line setpoint, 90R5, is not part of the DC SVA. It is a fixed value and only may be varied by adjusting the potentiometer manually. It is also used as a reference for field voltage during shutdown or after the line breaker trips. During start-up it is used as a reference until the turbine reaches 95% speed, at which time the AC regulator is placed in control.

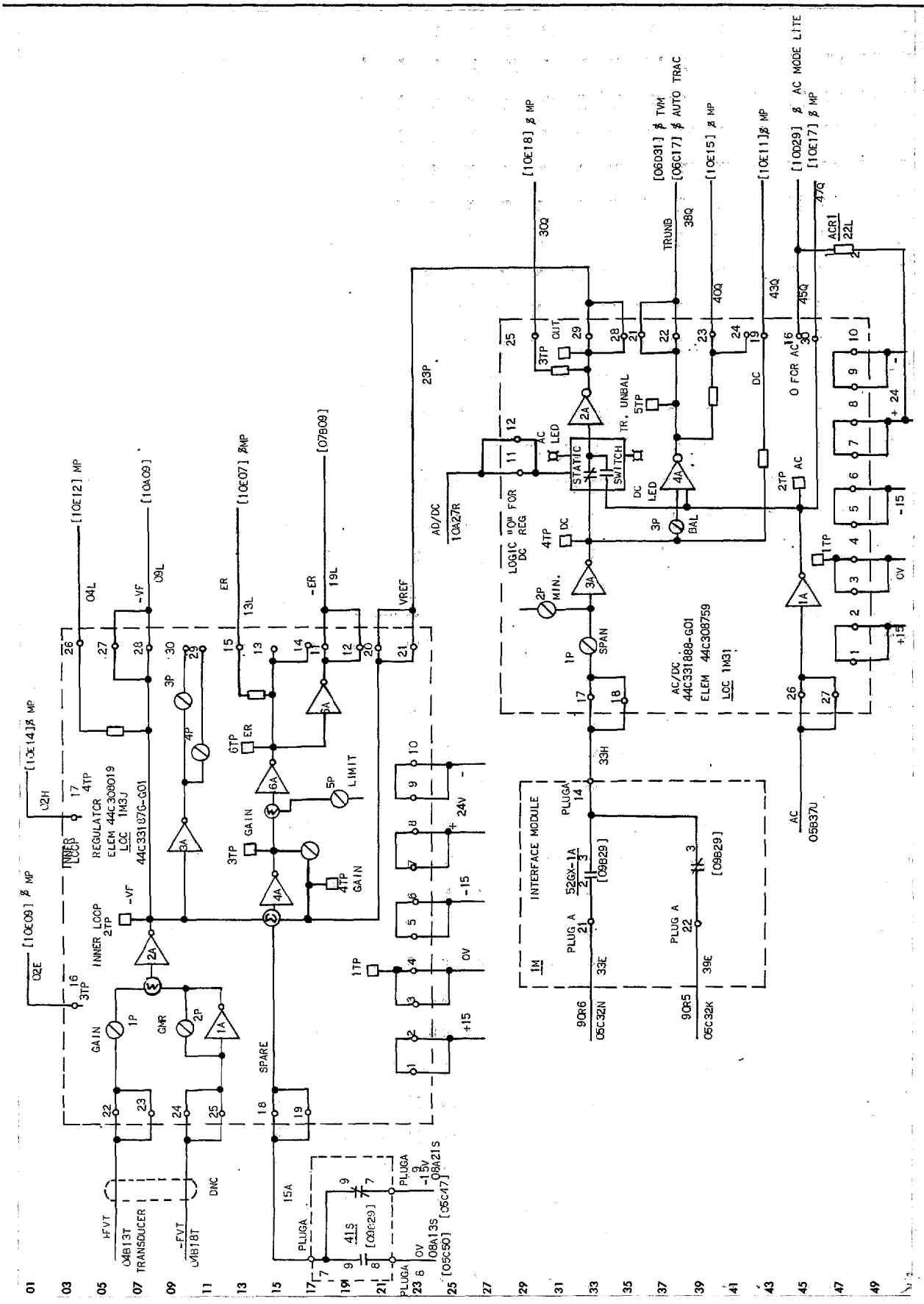


Figure 3
DC Regulator Elementary

During start-up, the DC SVA is adjusted to provide a set-point for the DC regulator, if it were placed in service while the generator is on-line. Normally, the DC Regulator on-line setpoint is adjusted by tracking the AC setpoint while the AC Regulator is functioning. Provisions must be made to set the on-line DC setpoint reference to .75 VFFL before the AC Regulator is placed in service. The DC SVA accomplishes this by using the tracking control signal and potentiometer 90R6. To enable this function to operate, relay 94Z-2 is energized during field flashing. This allows the output of the DC SVA to be compared with an on-line reference value set by 90R6. If the output is lower than the reference, the raise signal to the counter is given, raising the output. Thus while the 94Z-2 contacts are energized, the output of the DC SVA will be the same as 90R6 after the comparison logic has raised the output of the DC SVA to the on-line reference value. It will remain at this value until the AC Regulator mode takes control of regulation at 95% turbine speed. After 95% turbine speed, the output of the DC SVA will track the AC setpoint. If the Transfer and Tracking PWB were not functioning, the DC SVA would remain at the .75 VFFL setpoint. (Reference set by 90R6). The tracking capability is provided by the logic inputs and the Auto input to the DC SVA. The input signal to Auto on the DC SVA is the AC/DC signal from the Overvoltage PWB. The AC/DC signal is a signal dependent upon two factors: which regulator mode has been selected by the operator and if an overvoltage situation occurs. When in AC regulator mode, the voltage on the AC/DC signal line is +24 VDC. This signal, on the Auto input to the DC SVA, enables the tracking function. When in DC mode, the voltage on the AC/DC line is zero. If operating in AC regulator mode and overvoltage situation occurs, the voltage on the AC/DC line will go to zero volts and tracking will stop.

When field flashing has stopped, the system is transferred to AC Regulator mode, and the 94Z-2 coil becomes de-energized. The Track input on the DC SVA is disabled, leaving the Auto input enabled by the AC/DC signal from the Overvoltage PWB. This will allow the DC Regulator setpoint to track the AC regulator setpoint. The logic inputs to the DC SVA control the raising and lowering of the DC regulator setpoint. They are provided from the Transfer and Tracking PWB.

The Transfer and Tracking PWB provides logical raise and lower signals to the DC SVA. The outputs are determined by the polarity of the input signal. The input signal is the transfer error signal, TRUNB, generated on the AC/DC PWB. Its value ranges from +10 VDC to -10 VDC. If the DC Regulator reference is greater than the AC Regulator reference, the transfer error signal from the AC/DC PWB is negative. Conversely, when the DC Regulator reference is less than the AC Regulator reference the transfer error signal is positive.

From Figure 4, it can be seen that if the DC output is greater than the AC, the Lower LED will light and a time delay will start. When the time delay is complete, the lower signal is sent to the DC SVA Control PWB. If the AC output is greater than the DC, the RAISE LED will light and the time delay will again start. When the time delay is complete, the raise signal is sent to the DC SVA Control PWB.

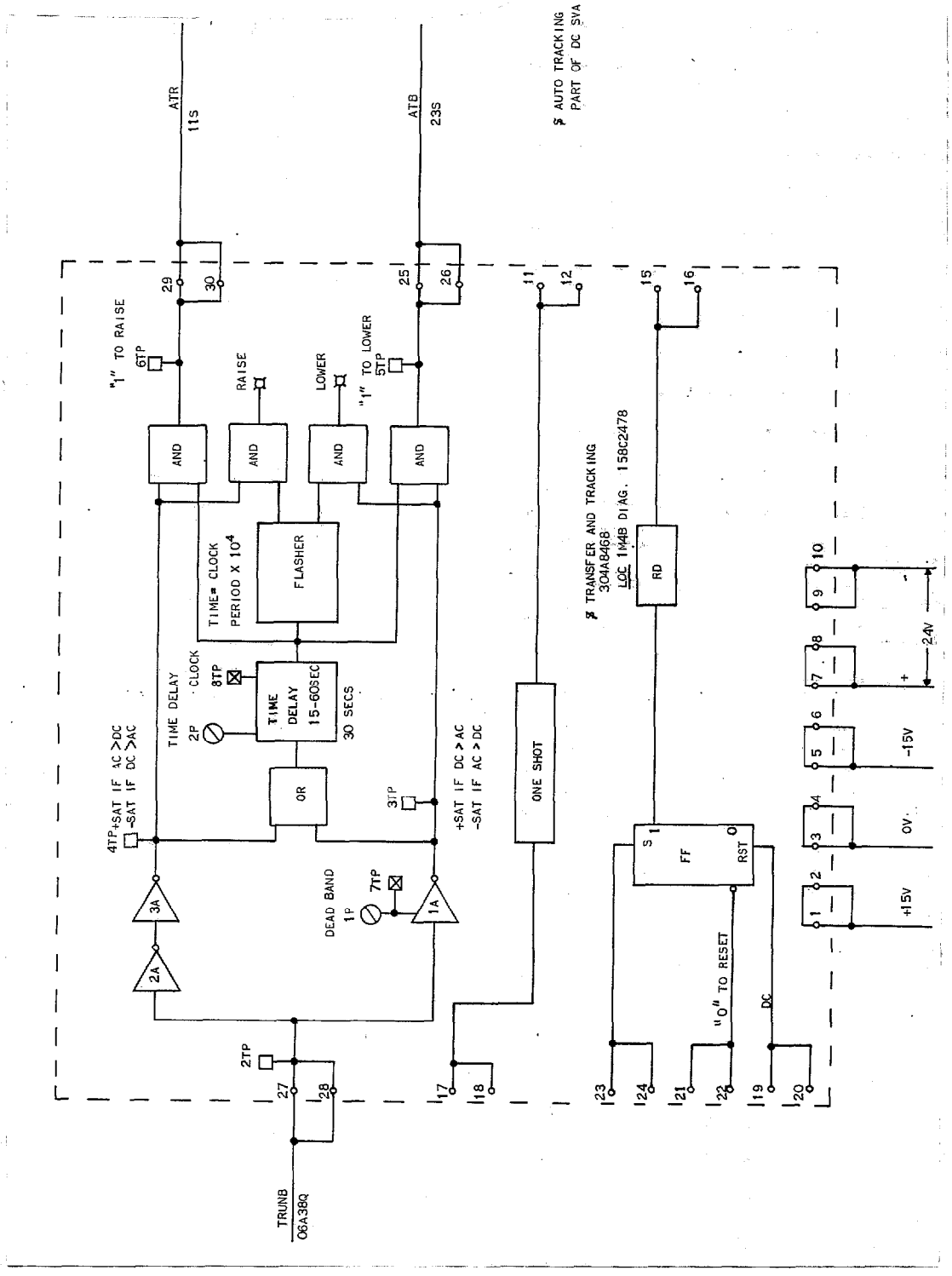


Figure 4
Transfer and Tracking Circuitry

The potentiometer 1P sets the deadband where the Transfer and Tracking PWB does not operate. The clock for the counter is generated from an oscillator on the board. After a total of 10,000 clock cycles have been counted, a logic 1 will appear at either the raise or lower output. The time delay, in seconds, from the start of the binary counter until a logic 1 is obtained at either the Raise or Lower output, is approximately ten times the period of the clock measured in milliseconds. This time delay is adjustable from 15 to 60 seconds. Provided on the Tracking and Tracking PWB are two LEDs, one to indicate a Raise output, and the other to indicate a Lower output. The Raise and Lower LEDs in the faceplate are on continuously for corresponding input errors and at the expiration of the time delay, the lamps will flash.

When a manual transfer from AC to DC Regulator or from DC to AC Regulator is to be initiated by the operator, an indication of the voltage difference in setpoints is required in order to assure a smooth transfer. This indication is provided by the Transfer Voltmeter. The transfer voltmeter is a nulling type meter that will indicate a difference in the AC and DC Regulator setpoints. The voltmeter input is the transfer error signal (TRUNB) from the AC/DC PWB. When the DC Regulator setpoint is greater than the AC Regulator setpoint the voltmeter will indicate a negative error. When the DC Regulator setpoint is less than the AC Regulator setpoint, the voltmeter will indicate a positive error. When both setpoints are equal, a zero reading will appear on the meter.

In most applications, the Transfer Voltmeter will be long distances away from the excitation compartment. In order to assure that the signals are properly transmitted over these distances, a Volts-to-Milliamp converter is used to drive the Transfer Voltmeter. The Volt-to-Milliamp (V/Ma) Converter accepts the transfer error signal and converts it from a voltage level to a current signal. The transfer error signal varies from -10 volts to +10 volts, the respective V/Ma output is 4 milliamperes to 20 milliamperes respectively as shown in Figure 5.

Along with the Transfer Voltmeter indication, the operator is also provided with lights to indicate which regulation mode the system is operating in. This is provided through the 83RM relay shown in Figure 6. When the operator switches into the AC Regulator mode and the AC Regulator is operating properly, (no overexcitation alarm) the 83RM relay will energize and activate the AC light to assure the operator that the AC Regulator setpoint adjuster is controlling the terminal voltage.

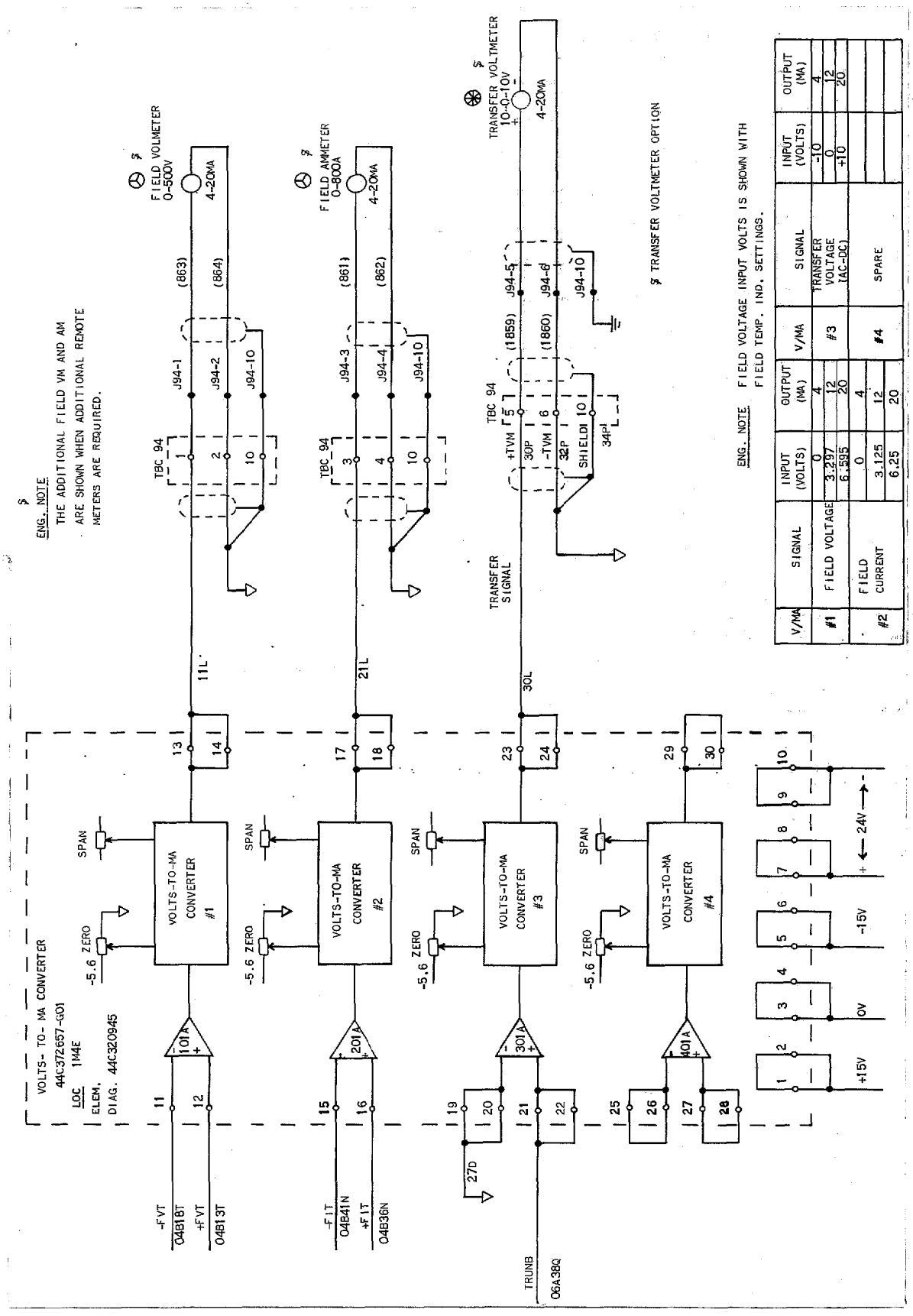


Figure 5
Volts-to-Milliamp Converter Elementary

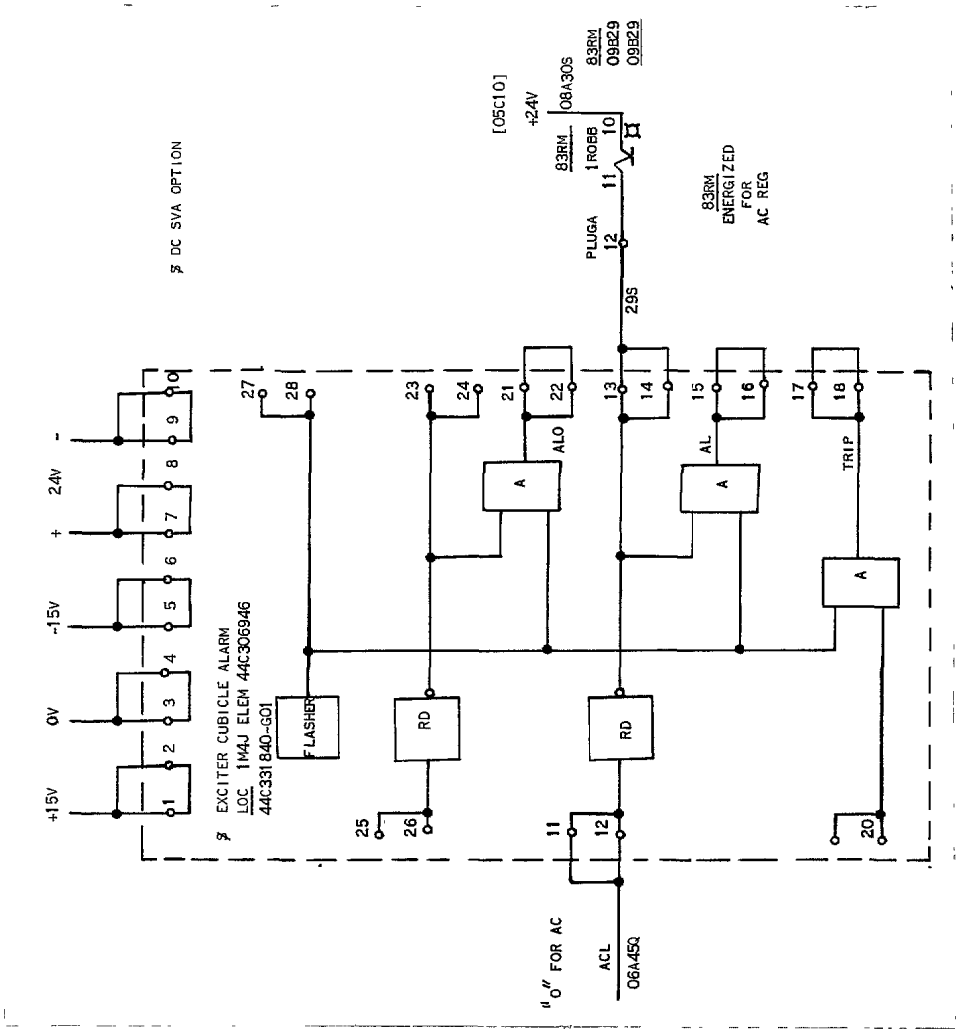


Figure 6
AC/DC Indication

INSTALLATION

The following procedures ensure that all the functions are working correctly. Since these functions are optional, the procedures must be interlaced with the standard set up procedures. For the DC Regulator controls, these procedures should be followed when references to the DC Regulator (90R5 or 90R6) are made.

Pre Excitation Checks

Static Voltage Adjuster (DC SVA) (Sheet 05C). Place the DC SVA Counter PWB on the card extender and monitor the SVA output voltage at test point 0 - (+) 10. Exercise the local RAISE pushbutton; the LSB (Least Significant Bit) LED should flash and the voltage increase. It should take approximately two minutes to reach the maximum limit of 10 volts. Adjust the GAIN potentiometer 1P for a +10.0 volt output if necessary.

Potentiometer 3P may be adjusted to change the time it takes the SVA counter to run from 0 to +10 VDC. This time span should be set to no less than 30 seconds since rapid changes may upset system stability. The nominal setting is 2 minutes from minimum to maximum. For a counting time of 2 minutes, the frequency measured at the clock test point on the SVA counter is approximately 35 Hz (28.0 millisecond period).

Reinsert the DC SVA Counter PWB into its location in the rack. Exercise the local LOWER pushbutton. The LSB indicator should light and the voltage at 0 - (+) 10 should decrease to zero. When power is removed from the SVA, the voltage is reset to zero. Check the operation of 90ML and 90MR by the remote Raise/Lower switch on the Generator Control Panel.

Lower the DC SVA to minimum position. Measure the voltage at 2TP (T-REF) of DC SVA Control board, which is dependent upon the setting of 90R6. Energize 94Z-2 while monitoring voltage at test point V-OUT on the SVA Control faceplate. The voltage at the test point should track the 90R6 setting as measured at T-REF. The Raise and Lower LEDs may both flash when the output matches. De-energize 94Z-2.

DC Regulator Set Points (Sheet 06A) (1M3I). Refer to Note E2 on Sheet A04 of the elementary for set point values for each adjustment.

CONTROL	FUNCTION
2P	Minimum Set Point
1P	Span Adjustment

Check the DC LED on the faceplate of the AC/DC board to ensure the DC regulator is in control of the excitation system. 83SR1 controls the AC or DC Mode.

NOTE: The following setting will be approximate as the regulator is not operating closed loop. The final settings will be performed OFF LINE.

1. MIN Set Point

   Measure the voltage at pins 17,18 of the AC/DC Board, and adjust the voltage to zero by 90R5. Measure the voltage at 2TP on the inner loop regulator PWB and set the MIN to the value on sheet A04, Note E2A, by adjusting 2P (MIN).

2. Span Set Point

   Energize 52GX-1A. Raise DC SVA to maximum limit. Measure the voltage at 2TP on the inner loop regulator PWB and set the span by adjusting 1P (SPAN) on the AC/DC faceplate for the value given on Sheet A04-E2B: de-energize 52GX-1A.

3. AC/DC PWB (1M3I, Sheet 06A)

   Remove the AC Regulator PWB from the rack assembly (the Field Overvoltage PWB must be in place). Jumper the AC SVA Counter output, pins 21,22 to pins 26,27 on the AC/DC board. (Adjust 90R5 for approximately -3.0). Adjust the SVA with the local controls so the voltage at 2TP is the same as that at 4TP. With the voltage of 2TP and 4TP equal, adjust potentiometer 3P (BALANCE) on the AC/DC PWB so that the voltage at 5TP is zero. Energize 83SR1. The DC LED will go out and the AC LED will light. The polarity of the voltage at 3TP will be opposite to that of 2TP, and will be the same as the output of the AC SVA.

At this time, verify proper operation of the Transfer Volt Meter (TVM) located at the generator panel. Measure the voltage at 5TP on the AC/DC PWB. The TVM should track 5TP as follows (See Sheet 06D for calibration):

1. 5TP = 0 Volts ; TVM should be on zero. MA out = 12.0
   AC(2TP) = DC (4TP)

2. 5TP = +10 Volts ; TVM should deflect to the right of zero.
   AC(2TP) >DC(4TP) MA out = 20

3. 5TP = -10 volts ; TVM should deflect to the left of zero.
   AC(2TP) <DC(4TP) MA out = 4

If TVM does not work properly, check for mis-wiring or mis-calibration of Volt-MA Converter board. Refer below for calibration information. Check the operation of 83RM (Sheet 10D elementary) and lights on GCP. De-energize 83SR1.

Volts/MA Converter Calibration.

1. Monitor the voltage at 5TP (TR UNBAL) on the AC/DC PWB. Adjust the AC SVA output simulating the AC Regulator output until the voltage at 5TP is 0.0 VDC.

2. Adjust the VOLTS/MA Converter #3 ZERO potentiometer to give a 12.0 milliampere output at HL15. Use the SPAN potentiometer to get close.

3. Measure the voltage at 5TP (TR UNBAL) on the AC/DC PWB. Adjust this voltage to +10.0 VDC using the AC SVA.

4. Adjust the VOLTS/MA converter #3 SPAN potentiometer to give a 20 milliampere output at TBC94-5.

5. Measure the voltage at 5TP on the AC/DC PWB. Adjust this value to -10 VDC by the the AC SVA output simulating the AC Regulator. The current at TBC 94-5 should be 4.0 milliamperes.

6. Repeat steps 3 through 5 until no further adjustment is required.

7. Note: At ±10 VDC (the TRUNBAL signal) the milliampere output need not be exact because the most significant point is when TR UNBAL is 0.0 VDC (12 milliamperes).

Automatic Tracking Adjustment (refer to Sheet 06C)

1. Place the Transfer and Tracking PWB on the extender in its location in the rack 1M4B.

2. Monitor the clock signal at 8TP with an oscilloscope. Adjust the period to 3.0 milliseconds using the TIME potentiometer 2P.

3. Close relay 83SR-1. The AC LED on the AC/DC PWB should be lit. Energize relay 52GX-1A.

4. Measure the voltage at 7TP. Adjust this value to 6.3 (±.1) VDC with the DEADBAND potentiometer 7P. This may be adjusted later.

5. Monitor the voltage at 5TP on the AC/DC PWB (marked TRUNB on the faceplate). This will be called the TRUNB signal.

6. Use the RAISE AND LOWER pushbuttons on the AC SVA Control PWB to adjust the TRUNB signal to a positive value.

7. Measure the voltage at 3TP on the Transfer and Tracking PWB. It should be -13 ±1.5 VDC. Measure the voltage at 4TP. It should be +10 (±.5) VDC. The RAISE LED will be lit.

8. Verify that after a time delay of 30 seconds, the RAISE LED will flash and the output of the DC SVA Counter PWB at the test point marked 0-(+)10 will increase. Make sure 94Z-2 is deenergized.

9. Lower the AC SVA setting with the LOWER pushbuttons until the TRUNB signal becomes negative.

10. Measure the voltage at 3TP. It should be +10(±.5) VDC. Measure the voltage at 4TP. It should be -13 (± 1.5) VDC. The LOWER LED will be lit.

11. Verify that after a time delay of 30 seconds, the LOWER LED will flash and the output of the DC SVA Counter PWB at the test point marked 0-(+)10 will decrease.

12. Remove the Transfer and Tracking PWB from the extender and place it in its location 1M4B.

13. When the Transfer and Tracking PWB is operating correctly, it may overshoot the AC setpoint slightly and then go the opposite way and again overshoot. This is OK. But to minimize the overshoot, adjust DEADBAND 1P CCW to decrease deadband. The Transfer and Tracking should be adjusted so that the TRUNB signal is within 1.0 volts of zero.

14. When checks are complete, remove the jumper from the AC SVA counter output to Pins 26,27 of AC/DC PWB. Replace the AC/DC PWB in the pack and drop out 52GX-1A.

Off-Line Set Point -90R5

Adjust 90R5 so the voltage at 2TP on the inner loop regulator PWB is the value shown on sheet A04, note E2C. This value is approximately equivalent to VFNL at 25°C. Potentiometer 90R5 should be locked at this setting until the generator is running off-line at no load. If required, 90R5 will then be adjusted to hold rated terminal voltage in the DC regulator mode.

On-Line Set Point - 90R6

Energize 52GX-1A. Measure the voltage at 2TP on the Inner Loop Regulator PWB. Adjust this voltage to the value given on sheet A04, note E2D with the DC SVA. Measure V_{out} on the DC SVA Control PWB and set the voltage at TREF (Same PWB) with 90R6 to the same value. Lock the knob.

Off-Line Checks

During off line operation, the DC Regulator is controlled by 90R5 and therefore there are no direct checks of the DC SVA.

While the AC Regulator is checked, ensure that the Transfer Voltmeter is working correctly.

After the normal off line checks are completed, perform the following checks.

Auto Tracking Checks

The Auto Tracking function can be exercised while off line by energizing 52GX-1A and selecting the AC Regulator (83SR1 energized).

When 52GX-1A is energized, 90R1 will cause an increase in generator voltage, the overexcitation setpoints will be changed to the on-line values, the DC SVA instead of 90R5 will control the DC Regulator and if the option is used, the VAR-PF controller will be enabled.

The DC SVA Output will track the output of the AC Regulator to keep the TVM nulled. The actual raise and lower signals will occur after a delay of 30 seconds. When the Raise/Lower signal stops, the TRUNB signal (5TP on AC/DC PWB will be within 1 volt of 0 VDC. Adjust the Transfer and Tracking deadband, potentiometer 1P to obtain close to this value.

While the DC SVA is tracking the AC Regulator, the DC Regulator set point adjuster should not have any control.

On Line Checks

Auto Tracking Checks. Verify the proper operation of the Auto Tracking function. As MW load and/or VAR load is raised, the field voltage will increase (in the AC Regulator). The Transfer Voltmeter will deflect to the right. After 30 seconds (from the initial point of operation) the Transfer Voltmeter will begin to move to the left. It should stop close to zero. During the time that the AC Regulator is in control, the AC Regulator set point adjuster will not affect the DC Regulator or Transfer Voltmeter.

When the Transfer Voltmeter is close to zero, a transfer to the DC Regulator can be made. If the AC/DC PWB and the Transfer and Tracking PWB are set up correctly, no appreciable change in field volts or VARS should occur. (The field voltmeter needle may jump slightly). Now the DC Regulator set point adjuster controls the excitation. The light controlled by 83RM should designate the control switch that the operator should use.

If all the checks above are satisfactory, no further adjustment of the Auto Tracking function are necessary.

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